L26_Lecture Scribe

L26_Stochastic Processes_...
PDF

Generate a lecture scribe suitable for exam preparation. While writing the scribe, explain the definitions and assumptions explicitly, give step-by-step reasoning for definitions, assumptions, proofs and the example step-by-step solutions from the provided context only. All the reasoning should be based solely on the provided context. Make the logical flow explicit so that a student can follow the reasoning during revision. Do not introduce material that is not present in the context.

Here is a **lecture scribe for exam preparation** based strictly on the provided lecture slides. Every definition, assumption, proof step, and example is explained **step-by-step with explicit reasoning**, following only the given content.

📘 Lecture 26: Stochastic Processes (SSS, WSS, Ergodic)

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1. Basics of Stochastic Processes

Definition

A **stochastic process** $X(\xi, t)$ is defined as:

- A **collection of waveforms/functions over time**
- Each waveform depends on the outcome ξ of a **random experiment**

Reasoning

- A random experiment produces outcomes → each outcome generates one waveform.
- Therefore, instead of one function, we get a **set of possible functions**.

Realizations and Random Variables

Key Concepts

1. **Realization**

- A **single observed waveform** from the process.

2. **Fix time** $t = t_1$:
   - $X(\xi, t_1) = X(t_1)$ becomes a **random variable (RV)**.
3. **Multiple time instances**:
   - $X(t_1), X(t_2)$ are **dependent random variables**.

Reasoning

- Fixing time → randomness remains → gives RV.
- Varying time → multiple RVs → their **dependency defines process behavior**.

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2. Characterization of Stochastic Processes

Probability Density Functions (PDFs)

1st Order

- Describes behavior at **one time instant**:

$$f_X(x, t_1)$$

2nd Order

- Describes **joint behavior at two times**:

$$f_{X_1 X_2}(x_1, x_2, t_1, t_2)$$

nth Order

- Complete description:

$$f_{X_1 \ldots X_n}(x_1, \ldots, x_n, t_1, \ldots, t_n)$$

Reasoning

- Higher-order PDFs capture **more dependencies**.
- Full knowledge of all orders → complete process description.

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3. Statistical Moments

Mean (First Order Moment)

Definition

$$\mu_X(t) = E[X(t)]$$

Using PDF

$$\mu_X(t) = \int_{-\infty}^{\infty} x \cdot f_X(x,t)\, dx$$

Reasoning

- Expectation gives **average value at time** t.
- Since PDF may depend on time → mean is generally **time-varying**.

Autocorrelation Function

Definition

$$R_{XX}(t_1, t_2) = E[X(t_1)X^*(t_2)]$$

Using joint PDF

$$R_{XX}(t_1, t_2) = \iint x_1 x_2 f_{X_1 X_2}(x_1, x_2, t_1, t_2)\, dx_1 dx_2$$

Reasoning

- Measures **relationship between values at two times**.
- Important for **prediction and estimation**.

Properties of Autocorrelation

1. Conjugate Symmetry

$$R_{XX}(t_1, t_2) = R^*_{XX}(t_2, t_1)$$

Proof (step-by-step from slides)

1. Start:

$$R_{XX}(t_2, t_1) = E[X(t_2)X^*(t_1)]$$

2. Take conjugate:

$$(E[X(t_2)X^*(t_1)])^*$$

3. Rearrange:

$$= E[X^*(t_2)X(t_1)]$$

4. Final:

$$= E[X(t_1)X^*(t_2)] = R_{XX}(t_1, t_2)$$

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2. Positive Definiteness

$$\sum_{i=1}^{n}\sum_{j=1}^{n} a_i^* a_j R_{XX}(t_i, t_j) > 0$$

Reasoning

- Represents **quadratic form**.
- Must be positive for any non-zero vector → ensures valid correlation structure.

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4. Covariance Function

Definition

$$C_{XX}(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X^*(t_2) - \mu_X(t_2))]$$

Relation

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$$

Reasoning

- Removes mean → gives **pure dependence**.

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5. Positive Definite Matrices

Conditions

1. All eigenvalues > 0
2. All leading principal minors > 0

Example (from slides)

- Determinants:
  - $|A_1| = 2 > 0$
  - $|A_2| = 3 > 0$
  - $|A_3| = 4 > 0$

Conclusion

→ Matrix is **positive definite**

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6. Stationarity

Concept

A process is **stationary** if:

- Its statistical properties **do not change under time shift**

Strict Sense Stationary (SSS)

Definition

$$f_X(x_1, \ldots, x_n; t_1, \ldots, t_n) = f_X(x_1, \ldots, x_n; t_1 + \tau, \ldots, t_n + \tau)$$

Reasoning

- Entire **joint PDF is invariant**
- Must hold for **all orders** n → very strong condition

Key Result

From derivation:

$$R_{XX}(t_1, t_2) = R_{XX}(t_1 - t_2)$$

Reasoning

- Time shift removes dependence on absolute time
- Only **time difference matters**

Wide Sense Stationary (WSS)

Conditions

1. Mean is constant:

$$E[X(t)] = \mu_X = \text{constant}$$

2. Autocorrelation depends only on lag:

$$R_{XX}(t_1, t_2) = R_{XX}(t_1 - t_2)$$

WSS vs SSS

- SSS ⇒ WSS
- WSS ⇏ SSS

Special Case (Gaussian Process)

- If Gaussian:
  - Mean + covariance fully define process
  - Hence:

$$\text{WSS} \Rightarrow \text{SSS}$$

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7. Example 1: Proving WSS

Given

$$X(t) = A\cos(\lambda t) + B\sin(\lambda t)$$

Conditions:

- $E[A] = E[B] = 0$
- $E[A^2] = E[B^2]$
- $E[AB] = 0$

Step 1: Mean

$$E[X(t)] = E[A\cos(\lambda t) + B\sin(\lambda t)]$$

Using linearity:

$$= \cos(\lambda t)E[A] + \sin(\lambda t)E[B]$$

Substitute:

$$= 0$$

✔ Constant → Condition 1 satisfied

Step 2: Autocorrelation

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$

Substitute:

$$= E[(A\cos\lambda t_1 + B\sin\lambda t_1)(A\cos\lambda t_2 + B\sin\lambda t_2)]$$

Expand:

- Cross terms vanish (since $E[AB] = 0$)
- Equal variances used

$$= E[A^2](\cos\lambda t_1\cos\lambda t_2 + \sin\lambda t_1\sin\lambda t_2)$$

Use identity:

$$\cos a\cos b + \sin a\sin b = \cos(a-b)$$

$$= E[A^2]\cos\lambda(t_1 - t_2)$$

✔ Depends only on $t_1 - t_2$

→ Process is **WSS**

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8. Example 2: Random Phase Cosine

Given

$$V(t) = \cos(\omega t + \theta)$$

- $\theta \sim \text{Uniform}[-\pi, \pi]$

Step 1: Mean

$$E[V(t)] = \frac{1}{2\pi}\int_{-\pi}^{\pi}\cos(\omega t + \theta)d\theta$$

Integrate:

$$= \frac{1}{2\pi}[\sin(\omega t + \theta)]_{-\pi}^{\pi}$$

$$= 0$$

✔ Independent of time

Step 2: Second Moment

$$E[V^2(t)] = \frac{1}{2\pi}\int \cos^2(\omega t + \theta)d\theta$$

Use identity:

$$\cos^2 x = \frac{1+\cos(2x)}{2}$$

Integral gives:

$$= \frac{1}{2}$$

✔ Independent of time

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9. Ergodic Process

Definition

A process is **ergodic** if:

- **Time average = Ensemble average**

Reasoning

- Ideally:
  - Average over many realizations
- Practically:
  - Only **one realization available**

If:

- Single realization gives correct average
  → Process is ergodic

Time Averages

Mean

$$\langle X(t) \rangle = \lim_{T \to \infty} \frac{1}{T} \int_{-T}^{T} x(t)\, dt$$

Autocorrelation

$$R_{XX}(\tau) = \langle x(t) x(t+\tau) \rangle$$

Key Result

- If WSS + ergodic:
  - Time averages = statistical averages

🔚 Final Summary

- **Stochastic process** → collection of random waveforms
- **Mean & autocorrelation** → key statistical tools
- **SSS** → full distribution invariant
- **WSS** → only mean + autocorrelation invariant
- **Ergodic** → time averages = ensemble averages

If you want, I can convert this into a ↪ **last-day revision sheet** or ↪ **expected exam questions with answers** based strictly on this lecture.

Do you like this personality?